

ECE 5254

Power Converter Modeling and Control

Fall 2024

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Center for Power Electronics Systems

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Grading:

Homework (6) 90%

Class Interaction 10%

Homework:

Homework assignments are at the discretion of the instructor. All homework is due at the **11:30pm** on their due dates.

All the homework should be submitted into Canvas before due time. Late homework should be sent into TA's email (**10% penalty per day** after due time will be executed unless prior arrangements have been made with the instructor).

Honor Code:

You are expected to abide by the honor code in all work associated with this course. You are encouraged to discuss the course topics and homework problems with each other but all work that is turned in for a grade must be done on an individual basis.

The Honor Code pledge that each member of the university community agrees to abide by states:

“As a Hokie, I will conduct myself with honor and integrity at all times. I will not lie, cheat, or steal, nor will I accept the actions of those who do.”

Students enrolled in this course are responsible for abiding by the Honor Code. A student who has doubts about how the Honor Code applies to any assignment is responsible for obtaining specific guidance from the course instructor before submitting the assignment for evaluation. Ignorance of the rules does not exclude any member of the University community from the requirements and expectations of the Honor Code.

Course Software: SIMPLIS



Hints about installing and using SIMPLIS:

(1) A free copy of SIMetrix/SIMPLIS introduction version can be downloaded from:
<http://www.simplistechnologies.com/downloads/file/win32/simetrix-simplis>
(You may need to register to download)

(2) During installation, select SIMetrix/SIMPLIS. After creating a new circuit, select “File -> Select Simulator -> SIMPLIS”.

(3) Place the necessary components and probes through the “Place” menu. Setting initial conditions for the passive component in the circuit may be necessary (Snapshots from transient analysis can help to stabilize POP analysis).

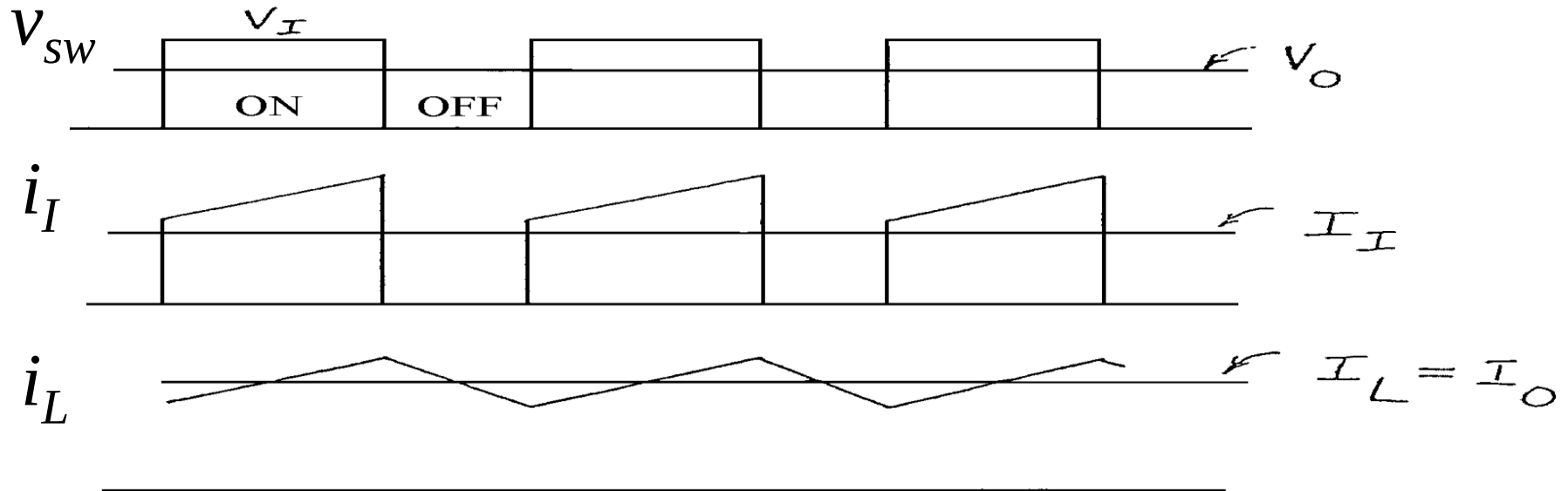
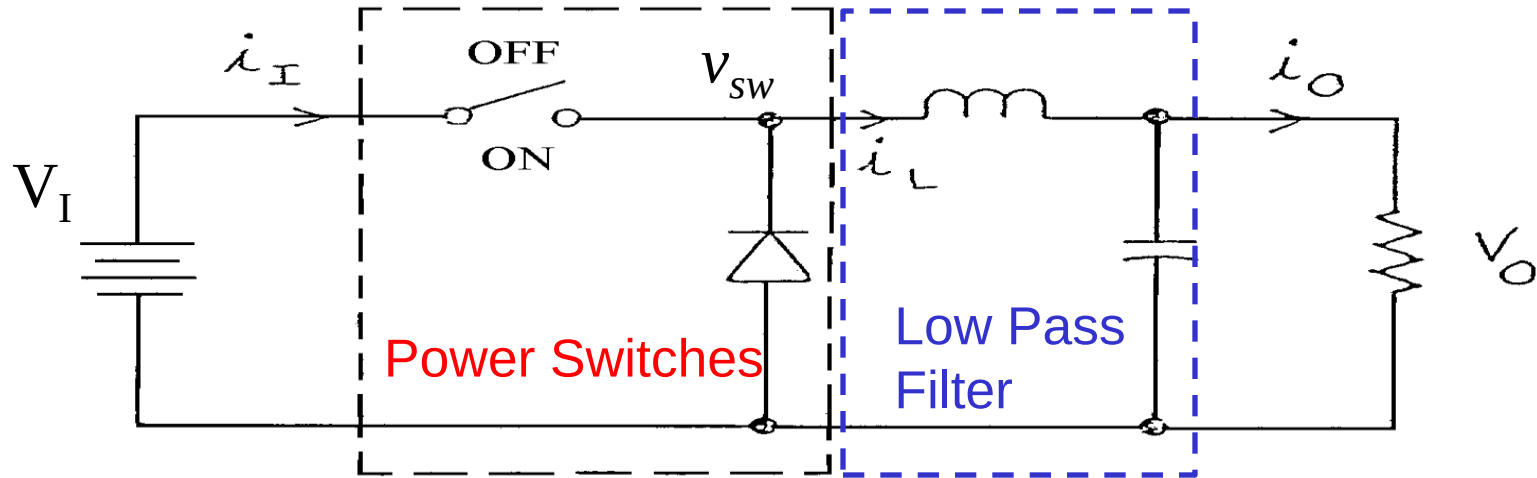
(4) The *“ideal diode in Simplis”* can be used as D or the anti-parallel diode. Change the parameters of the ideal diode in Simplis to make the forward voltage drop of the diode negligible.

(5) To enable POP function for a fast steady-state simulation and AC simulation, a POP trigger must be placed through “Place -> Analog Functions”. The input of this trigger should be connected to the switch gate signal.

(6) For AC simulation, select “Simulator -> Choose Analysis”. Enable AC function. The POP function will be enabled automatically.

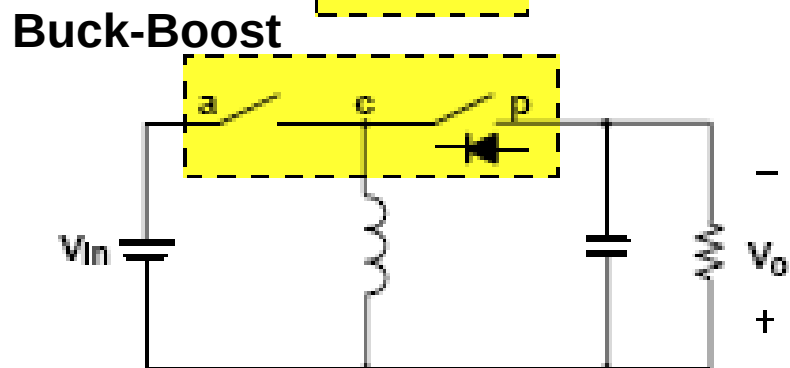
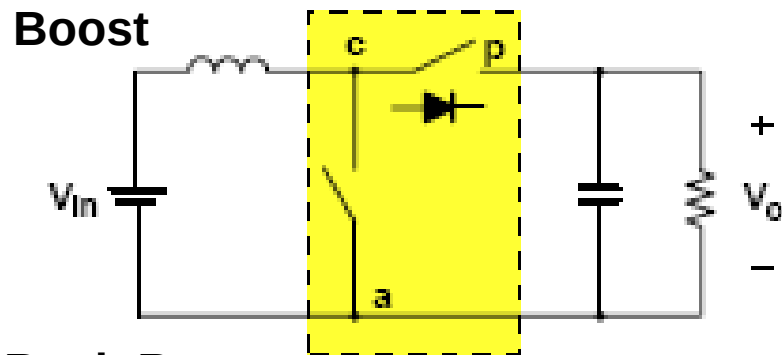
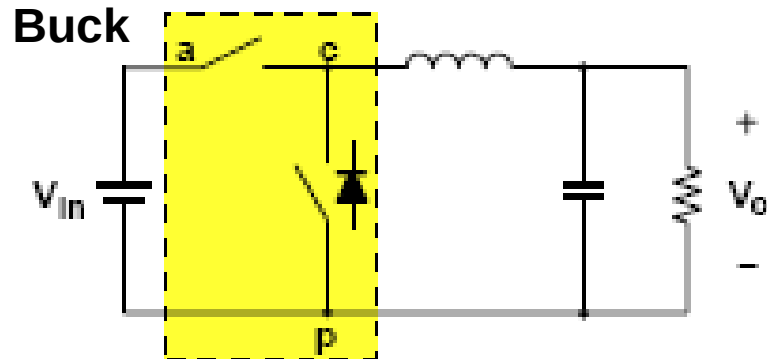
Pulse-Width-Modulation (PWM) Power Converters

BUCK CONVERTER



Issue with the PWM Power Conversion

--- *Non-linearity*



- ❖ Power Switches bring Non-linearity
- ❖ How to apply the linear control theory for analytical analysis?
- ❖ How to design the controller?
- ❖ How to predict the dynamic behavior?

Course Syllabus

- Chapter 1 — Average Model of PWM Converters**
- Chapter 2 — PWM Switch Model**
- Chapter 3 — Voltage-Mode Control**
- Chapter 4 — Peak-Current-Mode Control**
- Chapter 5 — New Current-Mode Control Modeling based on Describing Function Method**

Course Syllabus

Chapter 6 — Constant-on and V^2 Control

Chapter 7 — Average-Current-Model Control

Prerequisites: *ECE 4224* “Power Electronics” or similar course.

Understanding of basic power switching circuit techniques, as covered in 4224, is required before beginning the study of a more advanced power converter and control techniques.

Some comments from previous students...

What could you have done to be a better learner?

- Paid more attention in class
- spend more time on it and **more discussion**
- To prep a lesson before class
- **Read the course material before attending the classes**
- Before lecture, I could have taken more time to review the previous notes as well as skim through the upcoming notes to better prepare myself for the class. While I did this at the beginning of the semester, I started forgetting towards the end. **I noticed that when I reviewed the notes ahead of time, I was better able to follow along with the lectures.**